Participant-wise evaluation removes the apparent EEG signature of major depression

A pre-specified, leakage-free benchmark on the MODMA 128-channel dot-probe dataset, with a diagnosis of why published accuracies inflate

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Abstract

Background. Machine-learning studies often report near-perfect accuracy for detecting major depressive disorder (MDD) from EEG. Many of them, however, split trials rather than participants between training and test sets, and few compare against non-EEG baselines.
Methods. We re-analysed the 128-channel dot-probe ERP recordings of the MODMA dataset (24 MDD, 29 HC) from the raw files with a pre-specified, version-controlled plan. Preprocessing used PREP bad-channel detection and ICA with ICLabel, and participants were excluded by fixed quality criteria. We evaluated 18 model configurations:

- a demographic baseline;
- linear, kernel and tree models on 288 handcrafted features (spectral, frontal alpha asymmetry, ERP, wPLI connectivity, complexity);
- Riemannian models;
- three deep networks on full epochs.

All models used nested cross-validation over participants with identical folds. Results carry participant-level bootstrap confidence intervals, and all but the deep networks carry permutation tests of the full pipeline.
Results. In 45 participants, the pre-specified primary model (L2 logistic regression on all features) reached AUC 0.53 (95% CI 0.36–0.70, $p = 0.456$). No EEG model exceeded a baseline using only age, sex and education (AUC 0.78). The best secondary model (AUC 0.72) did not survive multiplicity correction, and nested model selection gave AUC 0.63. Evaluated with an invalid trial-wise split, the same tangent-space model achieved apparently perfect participant-level classification (AUC 1.00), versus 0.50 when participants were held out. The null result held when all 53 participants were included and when the analysis was restricted to a period in which both groups were recorded. Post-hoc exploratory analyses did not change it either:

- EEG added nothing to demographics (AUC 0.52);
- EEG with demographics regressed out inside each fold gave AUC 0.49;
- compact theory-driven features, a subject-level Riemannian model and reaction times were all at chance;
- the only borderline signal came from a random forest on raw ERP waveforms ($p = 0.049$, not significant after correction), which we propose as a hypothesis for independent testing.

Complexity and high-frequency features tracked a raw-data EMG index. With 18 patients, only large effects were detectable.
Conclusions. This study contributes a leakage-free reference benchmark for MODMA and identifies the mechanism behind inflated accuracies. A trial-level classifier recognised *which* participant produced a trial with 97.9% accuracy, so trial-wise splits reward person recognition rather than diagnosis. Evaluated on unseen participants, task EEG in this dataset carried no detectable diagnostic signal beyond a demographic baseline that reflects recruitment differences. Residual muscle activity is a concrete, checkable confound for complexity and high-frequency

features. Even an ideal score with a modest true effect (AUC 0.65) would need about 100 participants to be detected reliably. We provide a reporting checklist and open code so that future EEG studies of depression can be evaluated, and compared, on the same footing.

1 Introduction

Electroencephalography (EEG) is cheap, portable and widely available, which makes it an attractive candidate for objective markers of major depressive disorder (MDD). Many machine-learning studies report high accuracies for separating patients with MDD from healthy controls (HC) using EEG. However, a large share of this literature evaluates models by splitting *trials* or *segments*, rather than *participants*, between training and test sets.

When the same person contributes data to both sets, a classifier can succeed by recognising the individual (their head geometry, impedances, idiosyncratic rhythms or recording session) instead of the disorder. The resulting estimates do not approximate the intended use-case, which is diagnosing a new person [1]. This form of leakage is a documented driver of the reproducibility crisis in ML-based science [2]. It has recently been shown to inflate performance substantially in translational EEG studies [3].

Small samples aggravate the problem. With a few dozen participants, cross-validated accuracy has error bars of ± 10 percentage points or more [4], and theoretical chance levels are not attained in practice [5]. Finally, demographic or acquisition variables that differ between groups (sex, education, recording period) can masquerade as disorder-related signal.

The Multi-modal Open Dataset for Mental-disorder Analysis (MODMA) [6] provides 128-channel EEG recorded during an emotional-face dot-probe task [7] from patients with MDD and matched controls. It is a natural testbed for these questions. In this paper we ask a single question: *how well can task EEG classify MDD versus HC in participants that the model has never seen?*

What the reader gains. Beyond answering this question for MODMA, the study offers four things that can be reused:

- a *reference benchmark*: honest, participant-level estimates with fixed folds, against which future MODMA results can be compared;
- a *mechanistic explanation* of inflated accuracies, with the size of the inflation measured on the same data;
- *concrete confound checks*, namely a demographic baseline and a raw-data muscle index;
- *design guidance*: the effect sizes this sample can detect, the sample sizes needed for smaller effects, and a reporting checklist (Table 7).

Contributions.

1. A complete pipeline from the raw EGI files, with exact ID matching against the clinical spreadsheet, event decoding by channel name, per-participant preprocessing (PREP bad-channel detection, ICA with ICLabel) and pre-specified participant exclusion criteria.
2. An analysis plan committed to version control before any result existed. It fixes the primary model, the primary outcome and all thresholds, and every deviation is reported.
3. A strictly subject-wise evaluation of 18 model configurations across five families: a demographic baseline, handcrafted biomarkers (spectral, frontal alpha asymmetry, ERP, connectivity, complexity), Riemannian geometry and three deep networks.
    - Every model uses nested cross-validation and identical subject partitions.

- Every result carries a participant-level bootstrap confidence interval; all models except the deep networks also carry a label-permutation test of the full nested pipeline.

4. Confound analyses: can EEG predict sex, age or recording period, and does performance survive restriction to a period in which both groups were recorded?

5. A direct measurement of how much trial-wise splitting inflates performance on the same data, and its mechanism: single trials identify the participant almost perfectly.

6. Post-hoc audit analyses, labelled as exploratory:

   - the incremental value of EEG over demographics;
   - a raw-data EMG index;
   - power and sample-size estimates;
   - a feature dictionary documenting all 288 variables.

2 Related work

EEG markers of depression. Reviews of EEG markers of MDD have catalogued many candidates. These include quantitative band power, frontal alpha asymmetry (FAA), connectivity, vigilance, sleep EEG and event-related potentials. Effects are generally small and heterogeneous across studies, and the reviews repeatedly call for standardised procedures [8–10].

The most studied single marker is FAA [11]. A meta-analysis concluded that it has no diagnostic value for MDD [12].

Emotional ERPs and the dot-probe task. The dot-probe paradigm measures attentional bias towards emotional stimuli [7]. ERP studies of emotional faces typically examine three components:

- the face-sensitive N170 [13];
- the P300, indexing attentional allocation [14];
- the late positive potential (LPP), indexing sustained motivated attention [15].

The Lanzhou group that later released MODMA used a dot-probe task with emotional–neutral face pairs in 17 patients and 17 controls. They reported larger P300 amplitudes when the probe replaced a sad face, interpreted as difficulty disengaging from negative information, and explored ERP components as classification features [16]. The 128-channel ERP subset of MODMA (24 MDD, 29 HC) uses fear-, sad- and happy-neutral face pairs [6].

Machine learning on MODMA and on depression EEG in general. Studies applying machine learning to depression EEG often report accuracies above 90%. For example, a CNN–LSTM applied to EEG segments reached 97.7–99.1% [17].

Such figures depend critically on how data are split. When segments of the same participant appear in both training and test sets, a model can identify participants instead of the disorder. Segment-wise evaluation has been shown to overestimate performance in translational EEG [3], consistent with a broader literature on record-wise splitting and leakage [1, 2].

Fewer MODMA studies use subject-independent protocols. A graph-network approach on the 128-channel *resting* EEG, with leave-one-out validation, reported around 81% accuracy for multi-class severity recognition [18].

Published MODMA studies differ in signal type, preprocessing and validation, and often under-report whether folds are formed over participants. We therefore do not attempt a quantitative comparison. Instead, Section 4.3 measures directly, on the same data and pipeline, how much trial-wise splitting changes the estimate.

Table 1: Sample characteristics (all 53 recordings). Mean ± SD. †Descriptive only; never used as predictors.

Variable	MDD ($n = 24$)	HC ($n = 29$)	Test	p
Age (years)	30.9 ± 10.4	31.4 ± 9.1	Mann–Whitney	0.82
Education (years)	13.2 ± 3.8	16.0 ± 2.3	Mann–Whitney	0.0048
PHQ-9†	18.3 ± 3.5	2.7 ± 1.8	Mann–Whitney	<0.001
GAD-7†	13.4 ± 4.9	2.1 ± 2.1	Mann–Whitney	<0.001
PSQI†	12.5 ± 4.2	4.1 ± 2.2	Mann–Whitney	<0.001
Sex (F/M)	11/13	9/20	Fisher exact	0.39

Small samples. With $n \approx 50$, cross-validated estimates have wide confidence intervals [4]. Theoretical chance levels are also unreliable, so permutation-based significance testing is required [5]. Choosing the best of several models on the same cross-validation inflates its estimate [19]. These three facts motivate our pre-specified primary model, our participant-level bootstrap intervals, our full-pipeline permutation tests and our nested model-selection estimate.

3 Methods

3.1 Participants and data

We used the 128-channel event-related potential (ERP) recordings of MODMA (Lanzhou, 2014–2016) [6]. The release contains 53 recordings, one per participant, sampled at 250 Hz. They were acquired with a HydroCel Geodesic Sensor Net (EGI) using 128 scalp electrodes plus the vertex reference (E129/Cz).

According to the dataset description [6]:

- **Patients** were in- and outpatients of Lanzhou University Second Hospital. Each met DSM-IV criteria for major depression on a structured MINI interview, had a PHQ-9 score of at least 5 and had received no psychotropic drug treatment in the previous two weeks.
- **Controls** were recruited by posters and excluded if they had a personal or family history of mental disorders.
- **All participants** were 18–55 years old and had at least primary education.

The same participants also completed a resting-state recording, which we did not analyse. Medication history beyond the two-week washout, illness duration and comorbidity are not provided per participant.

Diagnostic group (24 MDD, 29 HC) was taken *exclusively* from the official subject-information spreadsheet. Recording files were matched to it by exact identifier, without normalisation: all 53 identifiers matched one-to-one.

Table 1 summarises the sample. The groups did not differ significantly in age ($p = 0.823$) or sex ($p = 0.394$), but MDD participants had fewer years of education ($p = 0.005$).

In the 45 analysed participants the education gap was larger:

- education: MDD 12.8 ± 3.5 vs. HC 16.0 ± 2.4 years, $p = 0.001$;
- age: 30.8 ± 10.7 vs. 31.3 ± 9.3 years, $p = 0.853$;
- sex (F/M): 8/10 vs. 8/19, $p = 0.354$.

Questionnaire scores (PHQ-9, GAD-7, CTQ-SF, LES, SSRS, PSQI) were used only to describe the sample and never as predictors, because they measure, or are closely tied to, the target.

Task and events. Each trial of the dot-probe task [7] consisted of the following sequence:

1. a fixation cross (464–652 ms);
2. a pair of faces, one emotional and one neutral (516 ms);
3. an inter-stimulus interval (116–316 ms);
4. a dot probe (168 ms), followed by the participant's response.

Trials came in three blocks of 160 (fear, happy, sad), with block order varying across participants.

We decoded events from the individual EGI event channels by name (`fcue`, `hcue`, `scue`). The combined stimulus channel produced by the reader carries the same code for every event in these files and is therefore unusable. Every recording contained the complete 480 trials. Epochs were locked to face-pair onset, because the probe appears at least 632 ms after it, so the window up to 600 ms is free of probe-evoked activity.

3.2 Pre-specified analysis plan

Before any epoch, feature or model output existed, we committed an analysis plan to version control. It fixed the preprocessing, the participant exclusion criteria, the feature definitions, the evaluation protocol, the primary model and the primary outcome.

Six deviations are listed in Appendix A:

- four were decided before any model was run;
- the fifth, a computational shortcut for the Riemannian models, before any of their results existed;
- the sixth adds post-hoc exploratory analyses after all results were known (Section 3.10).

The only one affecting the data was replacing an ad-hoc bad-channel rule with the PREP implementation (below), after the original rule produced implausible flags next to the reference electrode.

3.3 Preprocessing

All steps were performed separately for each participant with MNE-Python [20], so no information is shared across participants:

1. Bad channels were detected with the PREP pipeline (PyPREP `NoisyChannels`, without RANSAC) [21]. The criteria were NaN/flat signal, deviation, high-frequency noise, low correlation, low SNR, dropout and abnormal PSD. The vertex reference was exempt, being flat by construction.
2. Independent component analysis (Picard, extended-infomax objective [22], 30 components) was fitted on a copy filtered at 1–100 Hz, notched at 50 Hz (the mains frequency in China) and average-referenced, with bad channels excluded.
3. Components were labelled with ICLabel [23, 24]. A component was removed when its most probable class was eye, muscle, heart, line noise or channel noise *and* that class had probability ≥ 0.80.
4. The analysis copy was band-pass filtered at 0.5–40 Hz (zero-phase FIR) and notched at 50 Hz. It was then average-referenced, the ICA cleaning was applied, bad channels were interpolated (spherical splines), and it was re-referenced to the average.
5. Epochs spanned -200 to 600 ms, with a -200 to 0 ms baseline. An epoch was rejected if any channel exceeded $\pm 100\,\mu$V.

6. For spectral, connectivity and complexity features, the task period of the same cleaned recording was resampled to 125 Hz and cut into 2-s segments. The same amplitude rule was applied to the segments.

Participant exclusion (pre-specified). A participant was excluded if any of these held:

- more than 20% of channels were bad;
- fewer than 240 of 480 trials (50%) survived rejection;
- fewer than 40 trials survived in any condition.

3.4 Features

All features were computed per participant, without labels. They are therefore not a source of leakage; everything that *learns* from data happens inside the cross-validation loop.

Regions. We defined ten regions: frontal, central, temporal, parietal and occipital, each split into left and right (Fig. 10 in the appendix). Each HydroCel electrode was assigned to the region of its nearest 10–10 anchor position on the fitted head sphere. Midline electrodes and the lower face/neck ring (more than 115° from the vertex) were not used for regional features.

Regions differ in size. Central regions have 14 electrodes each, frontal 12–13, occipital 9, parietal 7–8 and temporal only 3. The temporal regions are therefore both the noisiest and the closest to the temporalis muscle.

Feature families.

- **Spectral (100 features).** Welch power in the delta (1–4 Hz), theta (4–8), alpha (8–13), beta (13–30) and low-gamma (30–40 Hz) bands, as $\log_{10}$ absolute and relative power per region.
- **Frontal alpha asymmetry (2).** $\ln \alpha_{\text{right}} - \ln \alpha_{\text{left}}$ for the HydroCel equivalents of F4/F3 (E124/E24) and F8/F7 (E122/E33) [11, 12].
- **ERP (21).** Measured per condition on the participant's average, using windows fixed a priori from the literature:
  - N170 peak amplitude and latency at left and right occipito-temporal clusters (around P7/P8), 140–200 ms [13];
  - P300 mean amplitude (250–500 ms) and peak latency at a centro-parietal cluster (around Pz) [14];
  - late positive potential (LPP), mean amplitude 400–600 ms, at the same cluster [15].
- **Connectivity (135).** Weighted phase-lag index (wPLI) [25] between the ten region-mean signals, in the theta, alpha and beta bands.
- **Complexity (30).** Per region, sample entropy [26], Higuchi fractal dimension [27] and Lempel–Ziv complexity [28], averaged over up to 100 2-s segments (250 samples). Parameters: sample entropy with $m = 2$ and $r = 0.2\,\text{SD}$; Higuchi with $k_{\max} = 10$; Lempel–Ziv on the median-binarised signal, normalised.

Peak latencies inside fixed windows are fragile. In 34% of participant–condition–side combinations the N170 minimum fell on the 200-ms window edge, and in 24% the P300 maximum fell on the 250-ms edge. The ERP latency features should therefore be read as low-validity measures. Appendix D describes every feature.

This gives 288 subject-level features. For trial-level models we kept the ten region-mean signals of each epoch (10×201 samples) and, for deep learning, the full 129×201 epochs.

3.5 Models

- **Demographic baseline.** L2 logistic regression on age, sex and education. It indicates how much of any EEG result could be explained by demographics alone.
- **Handcrafted features.** Each model is a scikit-learn [29] pipeline: median imputation, standardisation, then the classifier.
  - *Primary, fixed a priori:* L2-regularised logistic regression (LR-L2) on all 288 features.
  - *Secondary:* L1-LR, linear and RBF support-vector machines, random forest [30] and depth-limited LightGBM [31], plus LR-L2 on each feature family separately.

  Picking the best of many models inflates its estimate [19]. We therefore also report a *nested model selection* estimate, in which the inner loop chooses both the model family and its hyperparameters.
- **Riemannian, trial level.** [32, 33]
  - OAS-shrunk covariance matrices [34] of the regional epoch signals, either projected to the tangent space and classified with LR, or classified with minimum distance to mean (MDM);
  - ERP-augmented covariances, whose class prototypes are estimated inside each training fold, followed by tangent space and LR with a fixed $C = 0.1$ (deviation D5).
- **Deep learning, trial level.** EEGNet [35], ShallowFBCSPNet [36] and a temporal convolutional network (TCN) [37, 38], as implemented in braindecode.
  - Training used AdamW (learning rate 10^{-3}, weight decay 10^{-3}), cosine annealing, class-weighted cross-entropy, and augmentation by random time shifts ($\pm$40 ms), amplitude scaling and Gaussian noise.
  - Early stopping (patience 10, maximum 60 epochs) used the subject-level AUC on a validation split of the *training participants* (20%).
  - Channel-wise normalisation statistics came from the inner training participants only.

Trial-level models produce one probability per trial. These were averaged within each participant to give a single participant-level prediction.

3.6 Evaluation protocol

- **Outer loop.** Stratified 5-fold cross-validation over *participants*, repeated 10 times (3 times for deep learning, for computational reasons). All models were evaluated on *identical* participant partitions.
- **Inner loop.** Hyperparameters were selected by 5-fold participant-grouped cross-validation within the training participants, maximising AUC.
- **Leakage check.** An assertion verifies, in every outer, inner and validation split, that no participant contributes data to both sides.
- **Metrics.** All metrics are computed per participant: ROC AUC (the pre-specified primary outcome), balanced accuracy, sensitivity and specificity at a fixed 0.5 threshold (never tuned on test data), F1 and Brier score.
- **Uncertainty.** Point estimates are averaged over repetitions. 95% confidence intervals come from 2,000 stratified bootstrap resamples of *participants* applied to the out-of-fold predictions, with interval bounds averaged over repetitions.

- **Significance.** Label-permutation tests [5] permute labels *between participants* and re-run the entire nested pipeline. Fast models used 1,000 permutations; slower models used 200. We report $p = (1 + \#\{\mathrm{AUC}_{\mathrm{perm}} \geq \mathrm{AUC}_{\mathrm{obs}}\})/(1 + n_{\mathrm{perm}})$.

3.7 Confounds, sensitivity and inflation control

- **Can EEG predict confounds?** We applied the primary pipeline to three alternative targets: sex, age above the median, and recording period.
  - The recording period is relevant because group and period are partly confounded: between October 2015 and January 2016 only controls were recorded, and in March–April 2016 only patients.
  - We therefore also re-ran the primary pipeline on the participants recorded between June and August 2015, when both groups were recorded in parallel.
- **Exclusions.** Because exclusions were more frequent in MDD, we re-ran the primary model on all 53 participants.
- **Signal quality.** We compared retained trials, interpolated channels and removed ICA components between groups.
- **Inflation control (invalid by design).** To quantify the inflation caused by trial-wise splitting, we evaluated two trial-level models with both a trial-wise split and a subject-wise split, keeping everything else identical:
  - the tangent-space model;
  - a random forest on raw regional amplitudes, mimicking common practice.

3.8 Interpretation

- **Permutation importance.** For the primary model, computed on the *test* folds of the outer loop, for single features and for whole feature families permuted jointly.
- **SHAP.** SHAP values [39] of the primary model refitted on all participants (descriptive).
- **Univariate effects.** Cohen's d with Mann–Whitney tests and Benjamini–Hochberg false-discovery-rate correction [40].
- **Cluster-based permutation tests** [41] on channel-level band power (spatial clusters) and on ERPs (spatio-temporal clusters), as descriptive neurophysiology.

3.9 Data representations seen by each model

Table 2 lists the input of every model family. Every model was evaluated on the same participant partitions, and its result is always one prediction per participant.

Table 2: Inputs of each model family ($n = 45$ participants, 18,413 retained epochs).

Representation	Shape	Row	Models
Handcrafted features	45×288	participant	LR-L2 (primary), LR-L1, SVMs, RF, LightGBM, families
Regional epochs	$18{,}413 \times 10 \times 201$	trial	TS-LR, MDM, ERPCov-TS-LR, inflation control
Full epochs	$18{,}413 \times 129 \times 201$	trial	EEGNet, ShallowFBCSPNet, TCN
Demographics	45×3	participant	demographic baseline

3.10 Post-hoc audit analyses (exploratory)

After all pre-specified results were known, an internal audit added the following analyses (deviation D6). They are exploratory by construction. They do not replace or re-rank any pre-specified result, and the classification models among them are Holm-corrected among themselves.

- **Incremental value.** LR-L2 on demographics plus EEG features: all 288, or a compact set (below). The test permutes the rows of the EEG block between participants while labels and demographics stay fixed, so the null hypothesis is "EEG adds nothing to demographics".
- **Deconfounded EEG.** Within each training fold, the linear effect of age, sex and education was regressed out of every feature, and the training coefficients were applied to the test participants.
- **Dimensionality.** The 288 features are highly redundant: 53 pairs have $|r| > 0.9$, and 17 principal components explain 80% of the variance. We therefore added PCA fitted inside the training fold (5, 10 or 20 components, tuned in the inner loop) followed by LR.
- **Compact theory-driven set (36 features)**, fixed before it was evaluated:
  - ERP measures averaged over conditions;
  - sad−happy and fear−happy contrasts of N170, P300 and LPP amplitudes, because mood-congruent biases are differences between conditions;
  - the aperiodic ($1/f$) exponent and offset, obtained from a robust log–log fit at 2–35 Hz (the FOOOF logic, without a knee);
  - posterior alpha peak frequency and peak power above the aperiodic fit;
  - frontal-midline theta (Fz/FCz cluster), relative and aperiodic-corrected;
  - FAA;
  - trial-to-trial variability of the P300;
  - global wPLI per band, global relative band power and global complexity.
- **Behaviour.** The EGI files store event onsets but not trial keys, so congruency, side of the emotional face and accuracy are unavailable, and an attentional-bias index cannot be computed. Reaction times were derived as the first response 150–1500 ms after the probe: median, SD, coefficient of variation, miss rate, per-emotion medians and the sad−happy difference. This behavioural model uses no EEG and no questionnaire.
- **Subject-level Riemannian model.** The Riemannian mean of each participant's trial covariances, which is label-free, was projected to the tangent space fitted inside the training fold and classified with LR. This gives 45 rows instead of 18,413 correlated trials.
- **Confounds.**
  - Can EEG predict education above the median? This uses the primary pipeline with 1,000 permutations.
  - Signal quality was compared on all 53 participants, before exclusion.
  - An EMG index was computed per channel from the *raw* recordings (1-Hz high-pass, 50/100-Hz notch, average reference, task period, bad channels removed) as the log ratio of 55–95 Hz to 1–40 Hz power. Its median over 2-s segments was correlated with the features.
- **Power.** Two analyses:
  1. an ideal binormal score with 18 MDD and 27 HC, against the 95th percentile of the primary permutation null;

2. a semi-synthetic analysis. The real feature matrix had its rows shuffled to destroy any real association. A shift of d standard deviations in the MDD group was then added to 10 random features, and the full primary pipeline was run (200 simulations per d, one repetition each).

- **Learning curve.** The primary and demographic models were refitted on random stratified subsets of 12–36 training participants within the same outer folds.
- **Subject identification.** Multinomial LR on tangent-space single-trial covariances, predicting which of the 45 participants produced a trial, with a trial-wise 5-fold split.
- **Permutation test of the raw-amplitude random forest** (subject-wise, repetition 0 settings).

4 Results

4.1 Sample and signal quality

Of the 53 recordings, 45 met the pre-specified quality criteria (18 MDD, 27 HC). Exclusions affected 6 MDD and 2 HC participants (Fisher's exact $p = 0.12$). Among included participants, the median number of retained face-locked trials was 427 of 480 (range 265–471), the median number of interpolated channels was 7 (range 1–25) and the median number of removed ICA components was 7 of 30 (range 1–13). None of these quality measures differed between groups (Mann–Whitney $p = 0.36$, 0.32 and 0.92, respectively; Fig. 9). Any group difference in the features is therefore unlikely to reflect a gross difference in data quality.

4.2 Classification of MDD vs. HC

Table 3 and Fig. 1 summarise all models. All estimates are participant-level and out-of-sample.

Primary analysis. The pre-specified primary model (LR-L2 on all 288 features) did not separate the groups: AUC = 0.53 (95% CI 0.36–0.70), balanced accuracy = 0.54, permutation $p = 0.456$ (Fig. 2). The primary hypothesis, that task EEG features classify MDD above chance in unseen participants, is therefore not supported.

Demographic baseline. Age, sex and education alone gave AUC = 0.78 (0.63–0.91, $p = 0.002$). Of the three variables, only education differed between groups (Table 1), so it presumably carries most of this signal. This baseline outperformed the primary EEG model on the same folds (mean per-fold ΔAUC = -0.25, corrected resampled t-test $p = 0.019$; Table 4). It was not exceeded by any EEG model.

Secondary models. Of the secondary models, only L1-regularised logistic regression reached nominal significance: AUC = 0.72 (0.56–0.87), uncorrected permutation $p = 0.013$. This result does not survive Holm correction over the 13 secondary models with permutation tests ($p_{\text{Holm}} = 0.169$). Its advantage over the primary model on the same folds was also not significant (ΔAUC = $+0.16$, $p = 0.106$).

The nested model-selection estimate is the honest estimate of "choosing the best handcrafted-feature model", because the choice of model family is made inside the inner loop. It was lower: AUC = 0.63 (0.46–0.79, $p = 0.080$).

No single feature family was informative on its own. LR-L2 AUCs were:

- spectral and FAA: 0.51;
- ERP: 0.42;

Table 3: Participant-level out-of-sample performance (MDD vs. HC, $n = 45$). Point estimates are means over 10 repetitions of 5-fold cross-validation (3 for deep learning). 95% CIs come from a participant bootstrap. Sensitivity and specificity are at a fixed 0.5 threshold. p_{perm}: label-permutation test re-running the full nested pipeline (number of permutations in parentheses); "–" means not run. The primary model is in bold.

Model	AUC [95% CI]	Bal. acc. [95% CI]	Sens.	Spec.	Brier	p_{perm}
Demographic baseline						
Age + sex + education (LR)	0.78 [0.63, 0.91]	0.76 [0.63, 0.88]	0.72	0.80	0.205	0.002 (1000)
Handcrafted features						
LR-L1, all features	0.72 [0.56, 0.87]	0.66 [0.51, 0.79]	0.67	0.64	0.223	0.013 (1000)
Nested model selection, all features	0.63 [0.46, 0.79]	0.60 [0.46, 0.74]	0.62	0.57	0.253	0.080 (200)
Random forest, all features	0.55 [0.38, 0.72]	0.55 [0.40, 0.70]	0.52	0.59	0.253	0.348 (200)
LightGBM, all features	0.55 [0.37, 0.72]	0.55 [0.41, 0.69]	0.40	0.70	0.325	0.353 (200)
LR-L2, all features (primary)	0.53 [0.36, 0.70]	0.54 [0.39, 0.69]	0.46	0.62	0.259	0.456 (1000)
Linear SVM, all features	0.51 [0.34, 0.68]	0.55 [0.40, 0.69]	0.65	0.45	0.246	0.479 (1000)
RBF SVM, all features	0.43 [0.27, 0.61]	0.50 [0.39, 0.62]	0.76	0.25	0.269	0.572 (200)
Feature families						
LR-L2, complexity	0.55 [0.38, 0.72]	0.57 [0.43, 0.72]	0.58	0.57	0.288	0.705 (1000)
LR-L2, spectral + FAA	0.51 [0.34, 0.68]	0.54 [0.40, 0.69]	0.48	0.60	0.282	0.325 (1000)
LR-L2, wPLI connectivity	0.47 [0.30, 0.65]	0.46 [0.32, 0.61]	0.39	0.54	0.304	0.768 (1000)
LR-L2, ERP	0.42 [0.25, 0.59]	0.43 [0.28, 0.57]	0.36	0.50	0.319	0.720 (1000)
Riemannian (trial level)						
ERP covariances + TS + LR	0.61 [0.43, 0.78]	0.56 [0.42, 0.70]	0.44	0.67	0.246	0.159 (200)
Tangent space + LR	0.50 [0.32, 0.67]	0.47 [0.32, 0.61]	0.39	0.54	0.278	0.532 (200)
MDM	0.35 [0.20, 0.52]	0.39 [0.25, 0.53]	0.28	0.50	0.290	0.891 (200)
Deep learning (trial level)						
TCN	0.48 [0.30, 0.65]	0.48 [0.34, 0.63]	0.41	0.56	0.287	–
EEGNet	0.47 [0.30, 0.65]	0.46 [0.35, 0.58]	0.17	0.75	0.324	–
ShallowFBCSPNet	0.44 [0.27, 0.61]	0.48 [0.35, 0.62]	0.31	0.65	0.350	–

- connectivity: 0.47;
- complexity: 0.55.

All permutation p-values were ≥ 0.325. Non-linear models (RBF SVM, random forest, LightGBM) did not improve on linear ones.

Trial-level models. Neither the Riemannian nor the deep-learning models exploited the trial-level data better than the handcrafted features (Table 3).

- Riemannian: tangent space + LR, AUC = 0.50; MDM, 0.35; ERP covariances, 0.61 ($p = 0.159$).
- Deep learning: EEGNet, AUC = 0.47; ShallowFBCSPNet, 0.44; TCN, 0.48. All confidence intervals include 0.5.

The deep networks reached high validation AUCs in some folds, but test AUCs fluctuated widely across folds. This is the signature of fitting participant-specific rather than diagnosis-specific structure with only 36 training participants per fold.

4.3 How much does trial-wise splitting inflate performance?

Fig. 3 contrasts the same two trial-level models under a trial-wise split, where trials of a participant can fall in both training and test sets, and a subject-wise split.

- **Tangent-space model.** Trial-level AUC was 0.86 with the trial-wise split and 0.51 with the subject-wise split. Averaging trial probabilities per participant gave a participant-level AUC of 1.00 under the trial-wise split, i.e. perfect apparent diagnosis, versus 0.50 under the subject-wise split.

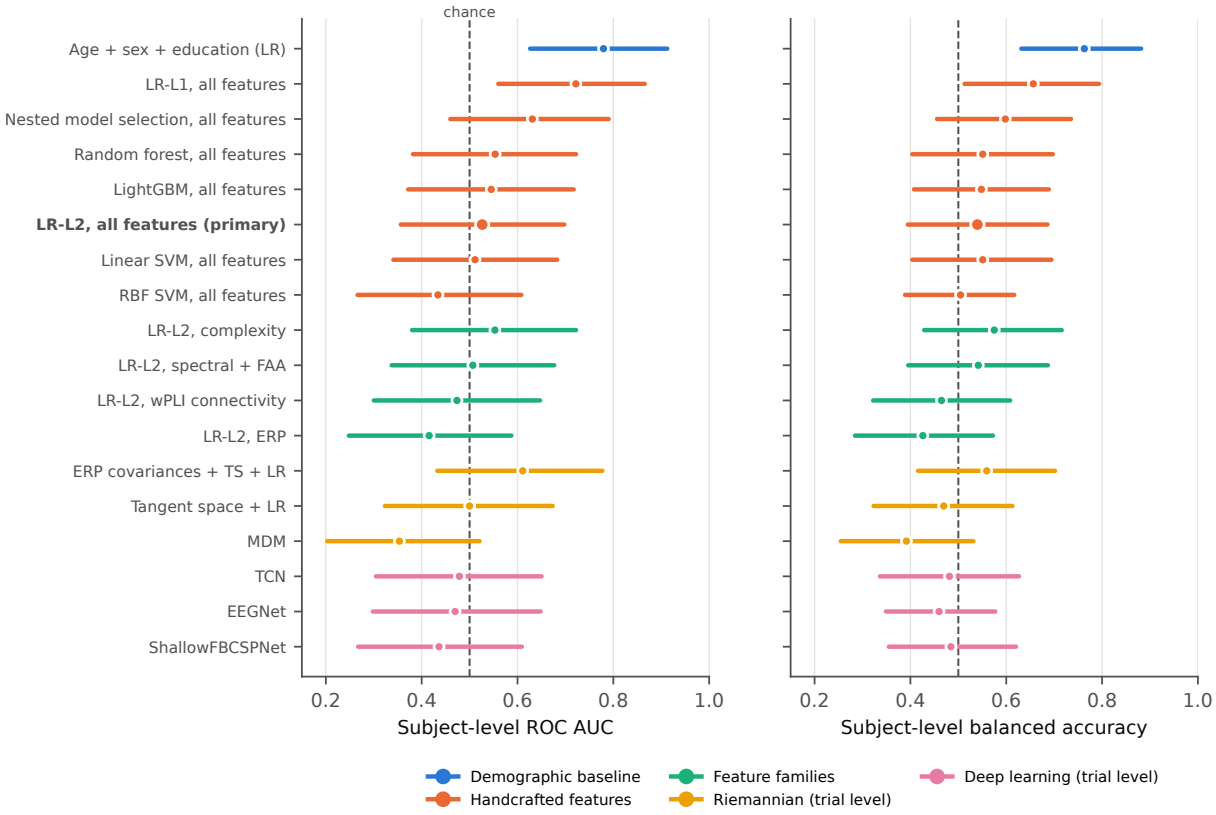


Figure 1: Participant-level ROC AUC (left) and balanced accuracy (right) with 95% bootstrap CIs, for every model. The dashed line marks chance. Only the demographic baseline (blue), which uses no EEG, has a CI clearly above chance.

- **Random forest on raw regional amplitudes**, the approach of many published studies and of an earlier unpublished analysis of these data by the author. Trial-level AUC was 0.77 versus 0.59, and participant-level AUC was 0.98 versus 0.70.

The trial-wise figures do not measure diagnosis. A classifier can recognise the *participant* from the individual statistics of their EEG, and the participant determines the label. Only the subject-wise figures answer the clinical question.

The subject-wise participant-level AUC of the raw-amplitude random forest (0.70) is the highest of any EEG model. We do not interpret it as evidence of a diagnostic signal, for three reasons:

- the model was included as a control, not pre-specified as a diagnostic model;
- it was evaluated with only three repetitions, and its post-hoc permutation test was borderline and did not survive multiplicity correction ($p = 0.049$, Holm $p = 0.390$; Section 4.6);
- it is one of many analyses, and for reference, 5% of permutation runs of the primary pipeline exceeded AUC 0.68.

It does suggest that the raw ERP waveform may be worth testing as a pre-specified hypothesis in independent data.

4.4 Confounds and sensitivity analyses

Table 5 reports whether the EEG features could predict variables other than diagnosis, using the primary pipeline.

- **Sex:** AUC = 0.63 (0.45–0.79, $p = 0.018$).

Table 4: Paired comparisons on identical participant folds. ΔAUC is the mean per-fold difference (model − reference). p_{corr} is from the corrected resampled t-test [42]. p_{Holm} applies Holm's correction to the permutation p-values of the secondary models.

	vs. primary		vs. demographics			
Model	ΔAUC	p_{corr}	ΔAUC	p_{corr}	p_{perm}	p_{Holm}
LR-L1, all features	+0.16	0.106	-0.09	0.470	0.013	0.169
Nested model selection, all features	+0.08	0.392	-0.17	0.176	0.080	0.955
Random forest, all features	+0.02	0.672	-0.22	0.029	0.348	1.000
LightGBM, all features	+0.01	0.944	-0.24	0.020	0.353	1.000
LR-L2, all features (primary)	–	–	-0.25	0.019	0.456	–
Linear SVM, all features	-0.13	0.422	-0.38	0.004	0.479	1.000
RBF SVM, all features	-0.13	0.429	-0.38	0.008	0.572	1.000
LR-L2, complexity	+0.06	0.517	-0.19	0.107	0.705	1.000
LR-L2, spectral + FAA	-0.01	0.903	-0.26	0.011	0.325	1.000
LR-L2, wPLI connectivity	-0.10	0.325	-0.34	0.008	0.768	1.000
LR-L2, ERP	-0.15	0.121	-0.40	0.000	0.720	1.000
ERP covariances + TS + LR	+0.08	0.455	-0.17	0.164	0.159	1.000
Tangent space + LR	-0.05	0.670	-0.30	0.027	0.532	1.000
MDM	-0.19	0.094	-0.44	0.000	0.891	1.000
TCN	-0.07	0.656	-0.31	0.078	–	–
EEGNet	-0.09	0.512	-0.33	0.022	–	–
ShallowFBCSPNet	-0.12	0.434	-0.36	0.025	–	–

Table 5: Can the EEG features predict confounds? The primary pipeline (LR-L2, all features) was applied to alternative targets, plus two sensitivity analyses of the MDD-vs-HC classification.

Target predicted from EEG features	n_{pos}/n	AUC [95% CI]	p_{perm}
sex (male)	29/45	0.63 [0.45, 0.79]	0.018
age > median	21/45	0.50 [0.33, 0.67]	0.617
recording period (>= Sep 2015)	18/45	0.55 [0.37, 0.72]	0.392
MDD vs. HC, primary pipeline, restricted subset			
Recorded Jun–Aug 2015	9/25	0.46 [0.24, 0.70]	0.517
All 53 participants (no exclusions)	24/53	0.57 [0.42, 0.72]	0.278

- **Age above the median:** 0.50 ($p = 0.617$).
- **Recording period:** 0.55 ($p = 0.392$).

EEG features thus carried information about sex, consistent with known sex differences in EEG amplitude and spectra. Sex was balanced across groups (Table 1), however, so it cannot explain a diagnostic signal, and none was found. Neither age nor recording period was predictable from EEG.

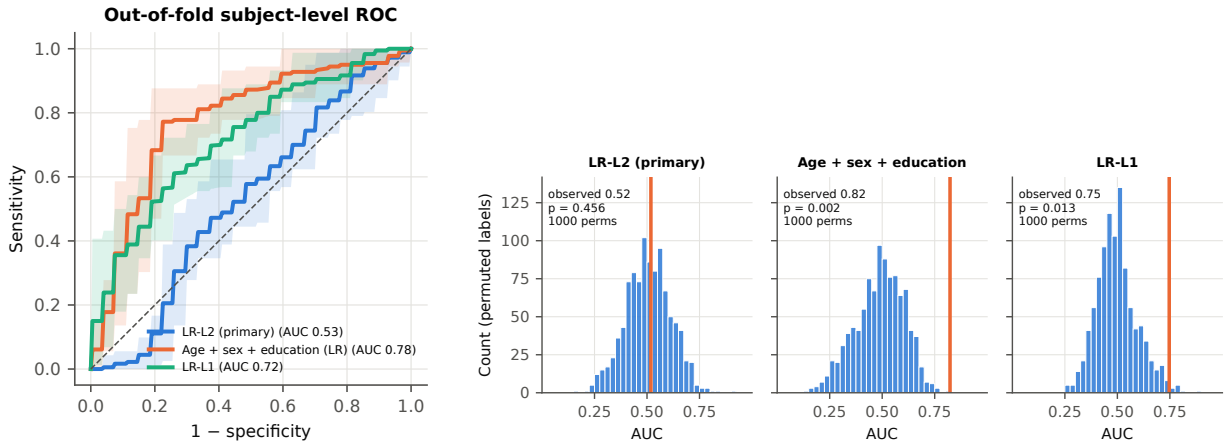
Restricting the analysis to the 25 participants recorded between June and August 2015 (9 MDD), when both groups were recorded in parallel, gave AUC = 0.46 (0.24–0.70, $p = 0.517$).

Including all 53 participants, with no exclusions, gave AUC = 0.57 (0.42–0.72, $p = 0.278$). The null result is therefore not an artefact of the exclusion criteria. Primary-model AUC was similar in women (0.56, $n = 16$) and men (0.51, $n = 29$).

4.5 What, if anything, differs between groups?

Model-based importance. Permuting whole feature families in the test folds of the primary model changed AUC by:

- complexity: 0.083;
- spectral: −0.001;
- ERP: −0.004;



(a) **(b)**

Figure 2: (a) Mean out-of-fold ROC curves across repetitions (band: 2.5–97.5% range over repetitions). (b) Permutation null distributions of the participant-level AUC, with the observed AUC (first repetition) marked.

- connectivity: −0.030.

All bootstrap intervals included zero (Fig. 4a). SHAP values of the refitted primary model (Fig. 4b) were small in absolute terms, consistent with the strong regularisation selected.

Univariate effects. 17 of 288 features differed between groups at uncorrected $p < 0.05$, close to the ≈ 14 expected by chance. Only 1 survived false-discovery-rate correction: theta-band wPLI between left central and left temporal regions ($d = 1.38$, $p_{\text{FDR}} = 0.036$; Table 8).

Spectral topography and ERPs. Fig. 5 suggests higher beta and low-gamma power in MDD, strongest at the edge of the montage, but no spatial cluster survived correction (minimum cluster $p = 0.06$).

Group-average ERPs (Fig. 6) showed the expected P1–N170 complex and a centro-parietal positivity in both groups. Visually, MDD participants had a smaller late positivity at occipito-temporal sites, but the spatio-temporal cluster test did not confirm any group difference (52 candidate clusters, minimum $p = 0.31$).

4.6 Post-hoc audit analyses (exploratory)

The analyses in this section were added after all pre-specified results were known (deviation D6; Section 3.10). Table 6 lists the classification models. None of them changes the conclusion.

Does EEG add to demographics? No.

- Adding all 288 EEG features to age, sex and education gave AUC = 0.52 (0.35–0.69). Permuting the EEG block gave $p = 0.746$.
- Adding the compact set gave AUC = 0.59 ($p = 0.584$).

Both are *below* the demographic baseline alone (0.78). With 45 participants, noisy EEG features dilute rather than complement the demographic signal.

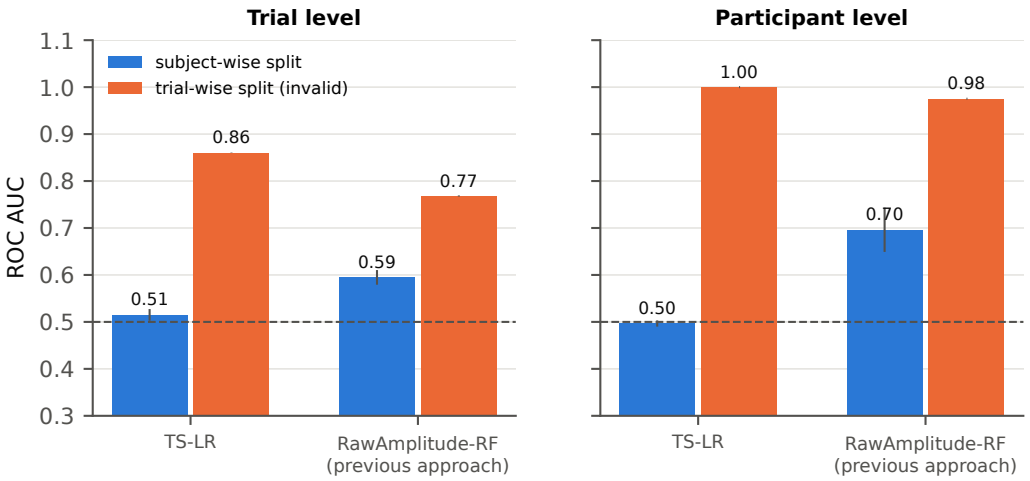


Figure 3: AUC of the same models evaluated with a trial-wise split (invalid; trials of each participant appear in training and test) and with a subject-wise split, at the trial level (left) and after averaging trial probabilities per participant (right). Bars show means over three repetitions, with SD as error bars. The trial-wise split is shown only to quantify the inflation.

Is the null result an artefact of confounding or redundancy? No.

- EEG with age, sex and education regressed out inside each fold: AUC $= 0.49$ ($p = 0.703$).
- PCA inside the fold: 0.58 ($p = 0.381$).
- The compact theory-driven set, including the ERP contrasts between emotions, the aperiodic exponent, the alpha peak frequency and frontal-midline theta: 0.51 ($p = 0.502$).
- Subject-level Riemannian model: 0.53 ($p = 0.064$).
- Reaction times alone: 0.44 ($p = 0.726$). Median RT was 312 ms in MDD and 304 ms in HC ($p = 0.754$).

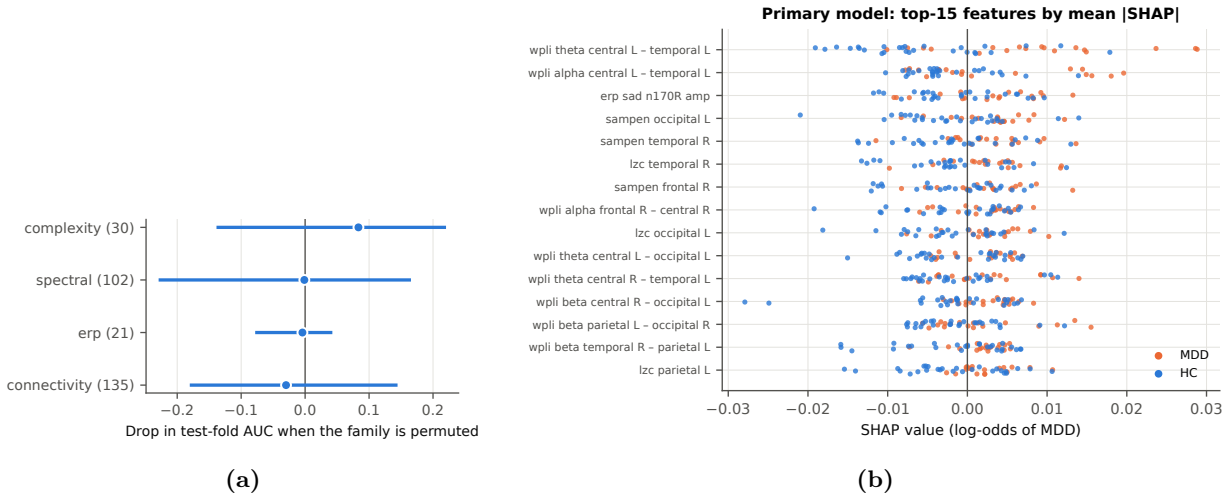
The smallest uncorrected p among these models was 0.064, and the smallest Holm-corrected p was 0.448.

Education as an EEG confound. The EEG features did not predict education above the median: AUC $= 0.66$ (0.42–0.87), $p = 0.190$. Only 9 of 288 features correlated with education at $p < 0.05$, about the number expected by chance, and 0 survived FDR correction.

Residual muscle activity. The EMG index, computed on the raw recordings, did not differ between groups ($p = 0.782$ in all 53 participants). It did, however, track the features that looked descriptively larger in MDD (Fig. 8):

- mean sample entropy, $\rho = 0.49$;
- mean Lempel–Ziv complexity, $\rho = 0.42$;
- relative beta power, $\rho = 0.42$;
- relative 30–40 Hz power, $\rho = 0.59$;
- aperiodic exponent, $\rho = -0.55$ (all $p \leq 0.004$).

These families therefore index muscle tone as much as brain activity. The aperiodic exponent was nominally lower in MDD ($p = 0.012$ in all 53; $p = 0.062$ in the 45 included). Given its correlation with the EMG index, this cannot be attributed to cortical excitation–inhibition balance. The FDR-surviving wPLI feature was not related to the EMG index ($\rho = 0.18$).



(a) **(b)**

Figure 4: (a) Drop in test-fold AUC of the primary model when each feature family is permuted jointly (mean and 95% interval over 50 outer folds; number of features in parentheses). (b) SHAP values of the primary model refitted on all participants, for the 15 features with the largest mean |SHAP| (descriptive).

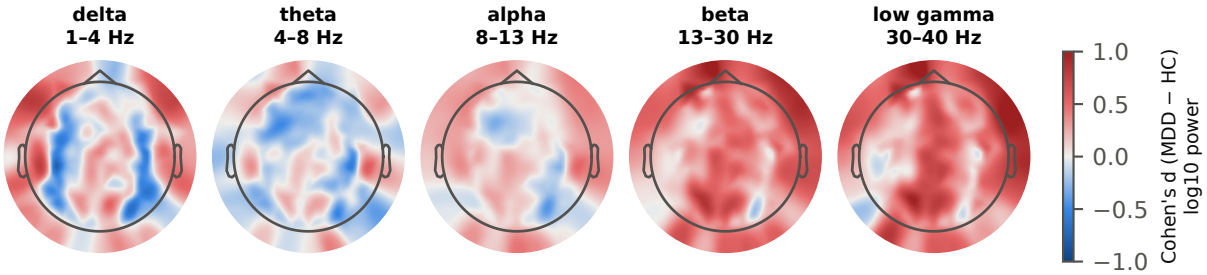


Figure 5: Group differences in log band power per electrode (Cohen's d, MDD − HC). Electrodes belonging to a significant cluster (cluster permutation test, $p < 0.05$) would be marked; none were.

Signal quality before exclusion. In all 53 participants, MDD and HC did not differ in:

- retained trials ($p = 0.362$);
- bad channels ($p = 0.316$);
- removed ICA components ($p = 0.921$) or ICLabel muscle components ($p = 0.589$).

Power. The 95th percentile of the primary permutation null was AUC 0.68. Even a perfect, already trained score would exceed it with probability:

- 61% at true AUC 0.70;
- 83% at 0.75;
- 95% at 0.80.

In the semi-synthetic analysis, where the primary pipeline has to learn the effect from 36 participants, power was lower (Fig. 7a):

- 14% when the injected signal had an oracle AUC of 0.83;
- 53% at oracle AUC 0.92;
- 82% at oracle AUC 0.97.

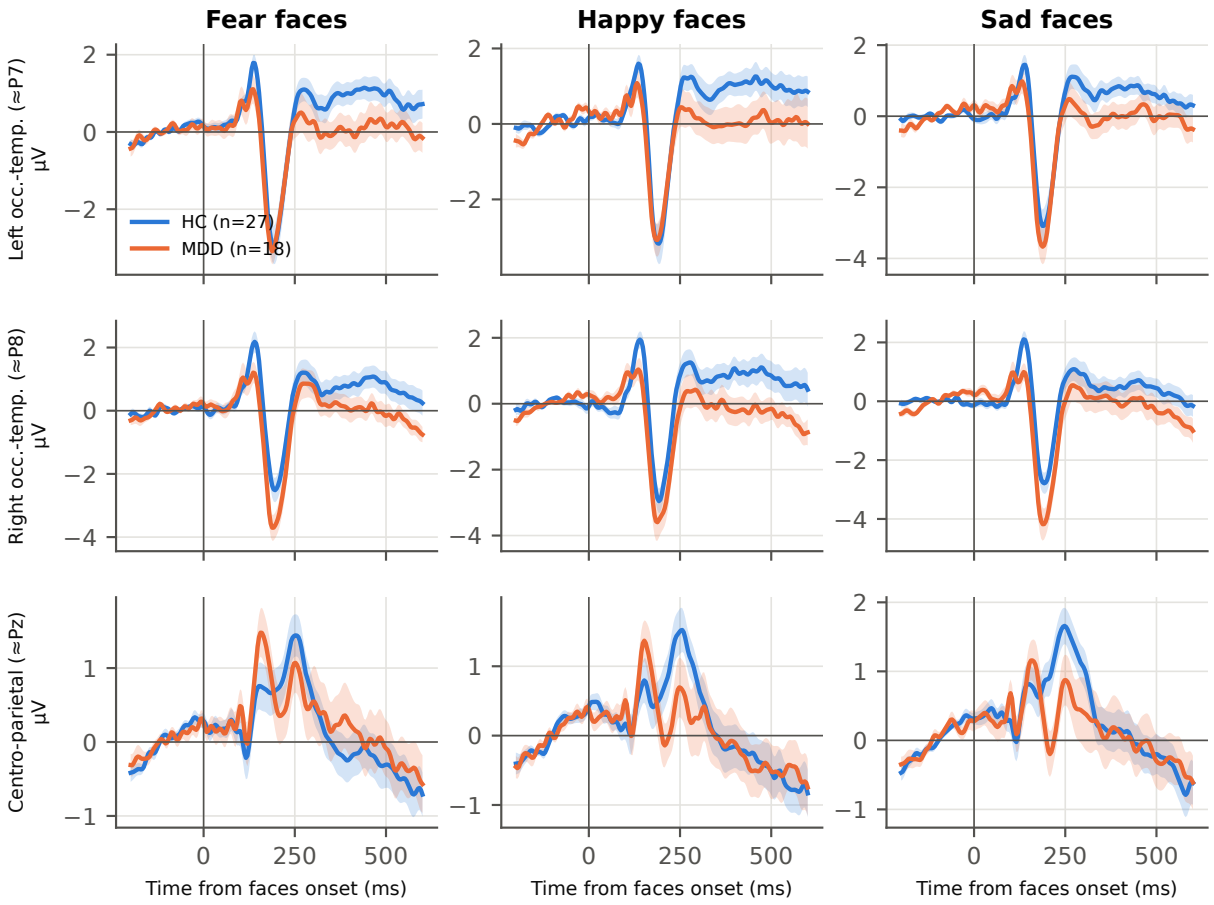


Figure 6: Grand-average ERPs (±SEM) per group and face condition, at the left and right occipito-temporal clusters and the centro-parietal cluster. Shaded windows are the a-priori N170 (140–200 ms) and P300 (250–500 ms) measurement windows.

The study could therefore only detect large effects. We extrapolated the same ideal score to larger samples, with the same class ratio and one-sided $\alpha = 0.05$. Reaching 80% power would need about:

- 60 participants for a true AUC of 0.70;
- 100 for 0.65;
- 300 for 0.60.

These are optimistic lower bounds, because a pipeline that also has to learn the effect needs more.

Learning curve. Between 12 and 36 training participants (Fig. 7b):

- the primary model moved from AUC 0.53 to 0.53;
- the demographic model moved from 0.72 to 0.78.

Why trial-wise splits inflate. A classifier on single-trial regional covariances identified *which* of the 45 participants produced a trial with 97.9% accuracy (chance 2.2%). Individual EEG signatures are thus highly distinctive, which explains the perfect participant-level AUC under trial-wise splitting (Section 4.3).

Table 6: Exploratory post-hoc models (D6), on the same folds as Table 3. p_{perm}: 1,000 permutations of the full nested pipeline. For the two incremental models the EEG block is permuted instead of the labels. p_{Holm} is computed across these seven models. ΔAUC is the difference from the demographic baseline, with the corrected resampled t-test.

Model (exploratory, post hoc)	AUC [95% CI]	BalAcc	p_{perm}	p_{Holm}	ΔAUC vs demo (p_{corr})
Demographics + all 288 EEG features	0.52 [0.35, 0.69]	0.54	0.746	1.000	-0.25 (0.014)
Demographics + compact EEG set	0.59 [0.42, 0.75]	0.59	0.584	1.000	-0.17 (0.089)
EEG 288, age/sex/education regressed out in-fold	0.49 [0.32, 0.66]	0.50	0.703	1.000	-0.28 (0.009)
EEG 288, PCA (in-fold) + LR	0.58 [0.42, 0.75]	0.58	0.381	1.000	-0.18 (0.085)
Compact theory-driven EEG set (36)	0.51 [0.34, 0.68]	0.53	0.502	1.000	-0.27 (0.009)
Reaction times only (8)	0.44 [0.27, 0.62]	0.49	0.726	1.000	-0.37 (0.002)
Subject-mean covariance, tangent space + LR	0.53 [0.35, 0.70]	0.49	0.064	0.448	-0.28 (0.065)

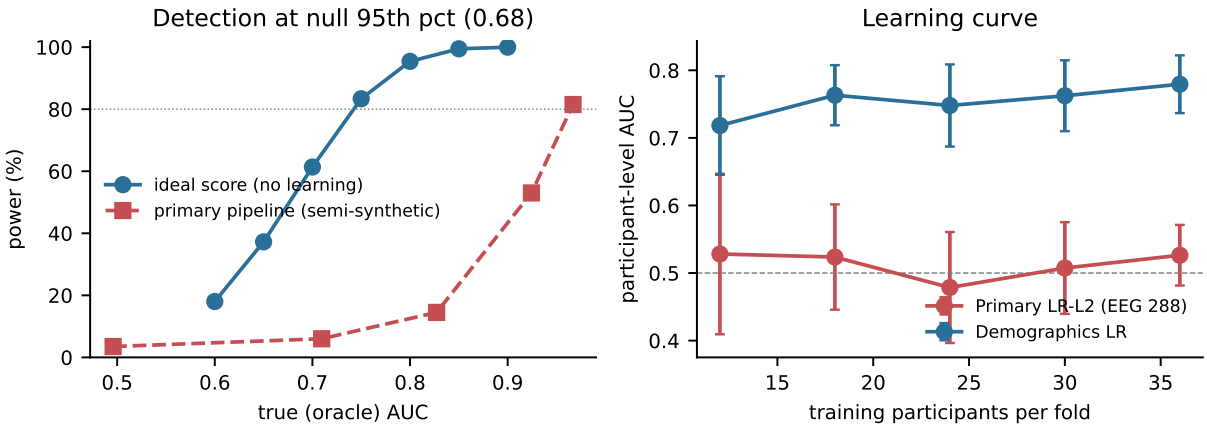


Figure 7: (a) Probability of exceeding the 95th percentile of the permutation null, as a function of the true AUC. Blue: an ideal pre-trained score. Red: the primary pipeline learning an injected effect, semi-synthetic. (b) Learning curve (mean ± SD over 10 repetitions).

Raw-amplitude random forest. With the settings of its first repetition (AUC 0.68), a permutation test of the full subject-wise pipeline gave $p = 0.049$. The test used 40 permutations, of which 1 reached the observed AUC, and the null 95th percentile was 0.61. This is the only nominally significant EEG result of the audit, but it is fragile:

- with so few permutations, the 95% interval for the exceedance probability extends to 0.13;
- after Holm correction across the 8 exploratory models it is $p = 0.390$;
- the model was not pre-specified, and a 288-feature linear model on summaries of the same data was at chance.

We therefore treat the raw ERP waveform as the most promising hypothesis this study generates, to be tested in independent data, and not as a finding.

5 Discussion

5.1 Main finding

Under a strictly participant-wise, pre-specified protocol, task EEG from the MODMA dot-probe recordings did not separate MDD from healthy controls better than chance. The conclusion holds across all model types tested:

- the pre-specified primary model and every feature family on its own;
- linear, kernel and tree-based classifiers;

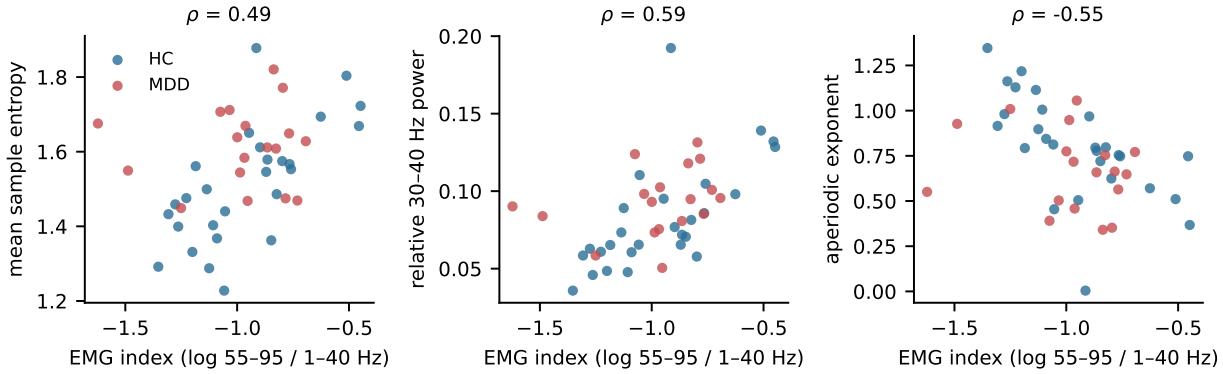


Figure 8: EMG index (raw recordings) against global sample entropy, relative 30–40 Hz power and the aperiodic exponent in the 45 included participants. ρ: Spearman correlation.

- Riemannian models and three deep networks trained on the full 128-channel epochs.

The one nominally significant secondary result (L1-regularised logistic regression) did not survive correction for the number of secondary models. The honest estimate of picking the best model, obtained by nested model selection, fell to 0.63, with a confidence interval including chance.

By contrast, three demographic variables (age, sex and education) reached AUC 0.78. This figure is not a diagnostic signal. It reflects how the sample was recruited: patients came from a hospital and controls from posters, and the groups differed by about three years of education. Any EEG model that claims diagnostic value on this dataset should still be benchmarked against it. The relevant question is whether EEG adds to demographics. In our exploratory analysis it did not: demographics plus EEG gave AUC 0.52 with all features and 0.59 with the compact set, both below demographics alone.

These findings agree with reviews describing EEG markers of depression as small and heterogeneous effects [8–10]. They also agree with the meta-analytic null result for frontal alpha asymmetry [12]. They do not agree with the near-perfect accuracies often reported for EEG-based depression detection [17].

5.2 Why do published accuracies differ so much?

Our inflation control offers a direct explanation. On the same data, the tangent-space model evaluated with a trial-wise split ranked every participant correctly (participant-level AUC 1.00; trial-level 0.86). Evaluated on held-out participants, it was at chance (0.50).

When epochs of one person appear in both training and test sets, the classifier only needs to recognise the person, and every person carries one label. Individual EEG signatures are highly distinctive, so this task is easy, and it says nothing about diagnosing a new patient [1, 3]. Reported accuracies therefore cannot be interpreted unless the split is stated to be across participants. In this light, values above 95% on datasets of a few dozen participants should be regarded with suspicion.

5.3 Model selection and multiplicity

With $n = 45$, sampling noise alone produces large AUCs: under label permutation, the primary pipeline exceeded AUC 0.68 in 5% of runs. Across 18 configurations, some will exceed that by chance. Reporting the best of them, here L1-LR at AUC 0.72, would have yielded a publishable but misleading result.

Three elements of the design guard against this:

- a primary model and outcome fixed before any result existed;

- multiplicity correction across secondary models;
- a nested estimate of the selection procedure itself [19].

We recommend all three as standard practice for small clinical EEG datasets.

5.4 Confounds

The groups differed in education and, partly, in recording period: some months contain only patients or only controls. In principle, period effects (equipment state, staff, season) could produce group differences unrelated to depression.

Two observations make this unlikely to be the explanation here. First, the EEG features did not predict recording period above chance. Second, the analysis restricted to the months in which both groups were recorded was no more successful than the full analysis.

The signal-quality measures did not differ between groups. Even so, one descriptive pattern deserves caution:

- MDD participants showed numerically higher beta and low-gamma power, strongest at peripheral electrodes;
- they also showed higher entropy and Lempel–Ziv complexity, the family whose permutation importance was largest, though not significant.

Both patterns are also typical signatures of residual muscle activity. ICA with ICLabel removes the clearest muscle components but not all EMG. The post-hoc EMG index supports this reading:

- it was computed on the raw recordings, before the 40-Hz low-pass;
- it correlated with entropy ($\rho = 0.49$), relative 30–40 Hz power ($\rho = 0.59$) and the aperiodic exponent ($\rho = -0.55$);
- it did not itself differ between groups.

Complexity, high-beta and aperiodic measures of task EEG should therefore be reported together with an EMG control before they are attributed to depression.

Education, the one demographic variable that differed between groups, is unlikely to explain the EEG results. The features did not predict it (AUC 0.66, $p = 0.190$), and regressing it out did not change the primary result (0.49).

5.5 A hypothesis worth testing

The one EEG model that came close to significance used the raw, regionally averaged ERP amplitudes of each trial, not summary features. Its participant-level AUC was 0.70 over three repetitions, and its permutation $p = 0.049$ did not survive correction (Holm $p = 0.390$). This pattern is consistent with a weak, temporally distributed ERP difference that hand-picked peak amplitudes and latencies miss. The latency boundary effects reported in Section 3.4 point the same way. A preregistered test of whole-waveform ERP models in an independent sample is the most concrete next step this study suggests.

The single connectivity feature that survived FDR correction (left central–temporal theta wPLI) was not robustly used by any model. It should be considered a hypothesis for replication, not a finding.

Table 7: Reporting checklist for EEG-based depression classification, derived from this study.

Check	Why it matters here
Split by participant, and say so	Trial-wise splitting gave AUC 1.00 versus 0.50 for the same model.
Fit everything inside the fold	Scaling, selection and tuning on the full data leak the test set.
Report a demographic baseline	Age, sex and education alone reached AUC 0.78.
Test the increment over that baseline	Adding EEG did not improve on demographics (0.52).
Permute the full pipeline	Chance AUCs up to 0.68 occurred at $n = 45$.
Correct for, or nest, model selection	The best of 18 models fell to 0.63 when its selection was nested.
Control for muscle activity	Complexity and gamma features tracked a raw EMG index (ρ up to 0.59).
Report participant-level CIs and power	With 18 patients, only large effects were detectable.
Pre-specify and log deviations	This keeps a negative result interpretable and a positive one credible.

5.6 What a negative result on MODMA is good for

A well-controlled null result is informative, for three reasons.

- **It is a benchmark.** Any new method evaluated on MODMA can be compared with our participant-wise estimates and fixed folds. A claim far above them should prompt a check of the split before it prompts excitement.
- **It explains the literature.** The near-perfect subject identification and the gap between trial-wise and participant-wise estimates show *why* high accuracies appear, not just that they do.
- **It sizes the next study.** The power analysis turns "larger samples are needed" into a number. Even under optimistic assumptions, a true AUC of 0.65 needs about 100 participants. That scale requires pooling across sites, which in turn makes site harmonisation and demographic baselines essential.

Table 7 condenses these lessons into checks that authors, reviewers and readers can apply to any EEG classification study of depression.

5.7 Limitations

- **Sample size.** With 18 MDD and 27 HC participants after exclusion, the participant-level confidence intervals are wide: for the primary model, 0.36–0.70. A modest true effect, around AUC 0.6, cannot be excluded. Our power analysis is explicit about this. Even an ideal pre-trained score with true AUC 0.70 would have passed the permutation threshold only 61% of the time, and a pipeline learning the effect from 36 participants needs an even larger effect. Our conclusion is that these data provide no evidence of a diagnostic signal, not that none exists.
- **Task information not available.** The released EGI files contain event onsets but not trial keys (congruency, side of the emotional face, accuracy). The attentional-bias index for which the dot-probe was designed, congruent-versus-incongruent probe ERPs and an exact replication of the probe-locked P300 effect of [16] could therefore not be computed. Reaction times alone did not separate the groups (AUC 0.44).
- **Task-state spectra.** Spectral, connectivity and complexity features came from the task recording. Most published spectral markers of depression are resting-state measures. The resting-state MODMA recordings of the same participants would allow a cross-paradigm check, but not an independent replication.

- **Single site and single task.** All data come from one site, one EEG system and one paradigm. Clinical details that may modulate EEG are not modelled, notably medication status and illness duration.
- **Asymmetric exclusions.** The pre-specified quality criteria excluded more patients than controls. The analysis on all 53 participants, however, gave the same conclusion.
- **Deep learning budget.** For computational reasons, the deep networks were evaluated with 3 rather than 10 repetitions and without permutation tests. Their confidence intervals all included chance, and their point estimates were below that of the primary model.
- **Feature space.** The features were fixed a priori from the literature. Other representations could perform better, but searching for them on the same 45 participants would reintroduce the selection bias this study was designed to avoid.

5.8 Conclusion

Evaluated rigorously on participants never seen by the model, 128-channel task EEG from the MODMA dot-probe dataset did not distinguish major depression from healthy controls, and did not add to a demographic baseline. The same data, evaluated trial-wise, yield apparently perfect diagnosis, because single trials identify the person. This contrast is the study's central lesson. It provides a quantified explanation for the gap between published accuracies and clinical reality, and an honest benchmark for this widely used dataset.

The path forward is clear and achievable:

- participant-wise, pre-specified evaluation;
- demographic and muscle-artefact controls;
- samples in the hundreds rather than dozens.

EEG may yet contribute to the assessment of depression, and studies built on these principles will be able to show it convincingly.

Code and data availability

All code, the analysis plan (fixed and version-controlled before any result was produced) and the list of deviations are available at `https://github.com/nicolasrodriguezco/eeg-mdd-pipeline`, with a companion book documenting code, methods and results at `https://nicolasrodriguezco.github.io/eeg-mdd-pipeline`. The EEG data belong to the MODMA project [6] and were obtained from its maintainers for academic use; they are not redistributed.

Acknowledgements

This work grew out of a research-seminar project at Universidad del Norte. The author thanks Dr. Karen Flórez for her supervision of that project, and the MODMA team for making the data available for academic research.

Use of AI tools

An AI coding assistant was used to write and test the analysis code and to draft the text. All analysis decisions, code and results were checked by the author, and every number in this paper is generated programmatically from saved outputs.

Disclaimer

This work is a methodological study. The models described here are *not* diagnostic tools and must not be used for clinical decision-making.

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A Deviations from the pre-specified plan

Deviations D1–D4 were decided before any model had been trained; at that point only the preprocessing log had been inspected. D5 was decided after the subject-level models had been evaluated but before any result of the affected (Riemannian) models existed.

D1 **Bad-channel detector.**

- *Problem:* the pre-specified rule (flat signal, robust log-SD $z > 5$, median neighbour correlation < 0.4) flagged up to 48 of 129 channels in a recording. Of all flags, 94% came from the neighbour-correlation criterion, and they included channels adjacent to the vertex reference with *negative* neighbour correlations. Whole-recording correlation is not a valid quality measure on reference-dominated, artefact-laden data.
- *Change:* the rule was replaced by the PREP implementation [21].
- *Unchanged:* the 20% exclusion threshold.

D2 **Re-run.** Fourteen participants were re-run after an infrastructure failure: a library was being installed during the first run. Since D1 required a full re-run anyway, all 53 were re-processed.

D3 **Added model.** The ERP-covariance Riemannian model was added as a secondary model.

D4 **Added sensitivity analysis.** Exclusions affected 6 MDD and 2 HC participants, so we added a sensitivity analysis of the primary model on all 53 participants.

D5 **Computational budget of the Riemannian models.**

- *Problem:* timing one outer fold showed that the full protocol (10 repetitions plus 200 permutations, i.e. 1,050 outer folds per model) would take more than 12 h, mostly because per-trial covariances were recomputed in every inner-loop fit.
- *Change 1 (no effect on results):* for the tangent-space and MDM models, per-trial OAS covariances were computed once. Each covariance depends only on its own trial, so this involves no learning across samples; the tangent-space reference point, scaling, classifier and MDM centroids were still fitted inside each training fold. A unit test verifies identical predictions.
- *Change 2:* the ERP-covariance model, whose class prototypes depend on labels and must be recomputed within each fold, used a fixed regularisation $C = 0.1$ instead of an inner grid. A value fixed in advance cannot inflate performance; at worst it is slightly sub-optimal.

D6 extbfPost-hoc audit analyses (exploratory). After all pre-specified results were known, an internal audit added the analyses of Sections efsec:audit-methods and efsec:audit. They do not replace, re-rank or re-threshold any pre-specified result. They are Holm-corrected among themselves and labelled exploratory throughout.

B Preprocessing quality

C Univariate group differences

D Feature dictionary

Every subject-level feature is computed per participant without labels. Names join the family, the band or measure and the region with underscores, for example `pow_alpha_frontal_L`. Regions are frontal, central, temporal, parietal and occipital, each left (L) or right (R).

Spectral, connectivity and complexity features are computed on the cleaned *task* recording, not at rest. Complexity and high-frequency power are sensitive to residual muscle activity (Section 4.6).

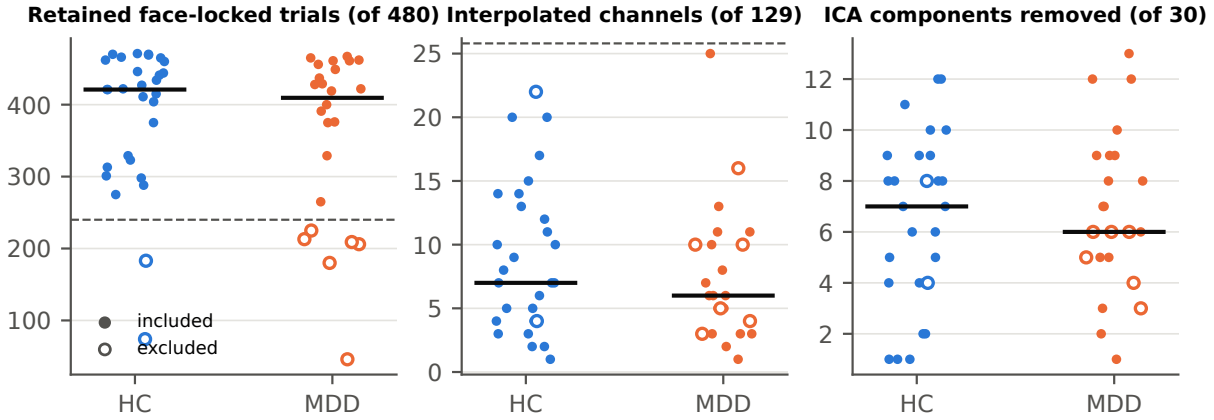


Figure 9: Signal quality per participant: retained face-locked trials, interpolated channels and removed ICA components. Open markers show participants excluded by the pre-specified criteria (dashed lines). Horizontal bars are group medians.

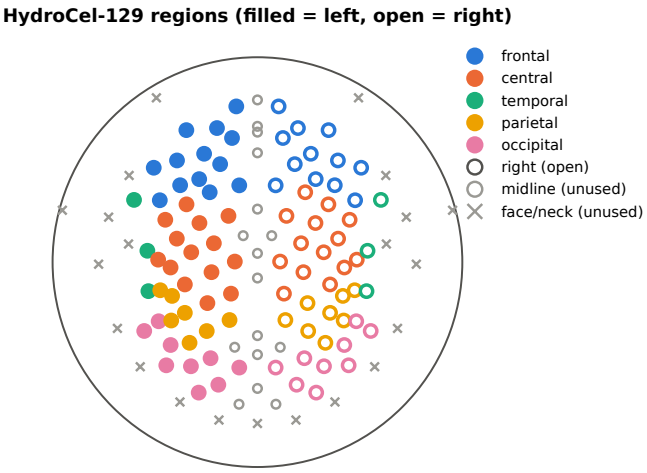


Figure 10: Assignment of HydroCel-129 electrodes to the ten regions, obtained by nearest 10–10 anchor on the fitted head sphere. Filled markers are left-hemisphere electrodes and open markers right-hemisphere electrodes. Grey electrodes (midline, and the face/neck ring more than 115° from the vertex) are not used for regional features.

Table 8: The ten features with the smallest uncorrected Mann–Whitney p (MDD vs. HC, included participants). p_{FDR} is Benjamini–Hochberg corrected over all 288 features.

Feature	Family	Cohen's d	p	p_{FDR}
`wpli_theta_central_L__temporal_L`	connectivity	1.38	<0.001	0.04
`sampen_occipital_L`	complexity	0.79	0.003	0.45
`rel_lowgamma_occipital_L`	spectral	0.57	0.005	0.47
`sampen_temporal_R`	complexity	0.76	0.009	0.62
`rel_beta_occipital_L`	spectral	0.69	0.012	0.66
`wpli_beta_central_R__occipital_L`	connectivity	-0.70	0.018	0.66
`sampen_parietal_L`	complexity	0.72	0.020	0.66
`lzc_occipital_L`	complexity	0.71	0.022	0.66
`lzc_temporal_R`	complexity	0.72	0.022	0.66
`pow_lowgamma_occipital_L`	spectral	0.64	0.024	0.66

Table 9: The 288 pre-specified subject-level features.

Name pattern	n	Definition and interpretation
`pow_<band>_<region>`	50	$\log_{10}$ absolute Welch power (µV^2) in delta 1–4, theta 4–8, alpha 8–13, beta 13–30 and low gamma 30–40 Hz, averaged over the region's electrodes. Depends on skull thickness and impedance as well as on brain activity.
`rel_<band>_<region>`	50	Band power divided by 1–40 Hz power. Removes individual scale. The five bands of a region sum to ≈ 0.94, so they are nearly collinear.
`faa_F3F4`, `faa_F7F8`	2	Frontal alpha asymmetry, $\ln \alpha_R - \ln \alpha_L$ (E124/E24, E122/E33). Lower values mean relatively less left-frontal activity, which approach–withdrawal theory predicts in depression.
`erp_<cond>_n170L/R_amp`, `_lat`	12	Minimum amplitude and its latency at 140–200 ms, at clusters around P7/P8, per condition (fear, happy, sad). Structural face encoding.
`erp_<cond>_p300_amp`, `_lat`	6	Mean amplitude (250–500 ms) and peak latency at a centro-parietal cluster around Pz. Attentional allocation.
`erp_<cond>_lpp_amp`	3	Mean amplitude at 400–600 ms, same cluster. Sustained motivated attention. Overlaps the P300 window ($r \approx 0.9$ with P300 amplitude).
`wpli_<band>_<A>__<B>`	135	Weighted phase-lag index between region-mean signals (45 pairs, theta, alpha and beta), multitaper over 2-s segments. It ignores zero-lag coupling, which reduces volume conduction.
`sampen_<region>`	10	Sample entropy ($m = 2$, $r = 0.2\,\mathrm{SD}$). Signal unpredictability.
`hfd_<region>`	10	Higuchi fractal dimension ($k_{\max} = 10$). Signal roughness.
`lzc_<region>`	10	Normalised Lempel–Ziv complexity of the median-binarised signal.